Finite Difference Analysis of a Rectangular Plate with Three Simply-Supported Sides and One Clamped Side
Under a Central Point Load

Theory of Plates Term Project

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Contents

1 Introduction

The aim is to:

- Derive the governing differential equation for plate bending.
- State the boundary conditions for simply supported and clamped edges.
- Discrete the governing equation using FDM on a uniform grid.
- Implement the formulation in Python and solve the resulting linear system.
- Perform a convergence study and discuss contour plots of deflection, bending moments, and shear forces.

2 Governing Equation of Plate Bending

2.1 Kirchhoff thin plate theory

The plate flexural rigidity D is

$$D = \frac{Eh^3}{12\left(1-\nu^2\right)}, \tag{1}$$

where E is Young's modulus and ν is Poisson's ratio.

The equilibrium of moments and forces in the plate leads to the biharmonic governing equation

$$D\nabla^4 w(x,y) = p(x,y), \tag{2}$$

where $p(x,y)$ is load per unit area (downwards) and

$$\nabla^4 = \nabla^2\left(\nabla^2\right) = \frac{\partial^4}{\partial x^4} + 2\frac{\partial^4}{\partial x^2 \partial y^2} + \frac{\partial^4}{\partial y^4} \tag{3}$$

is the biharmonic operator.

In this problem, the plate is loaded by a concentrated load P_0 at the centre $(x_c, y_c) = (a/2, b/2)$. In the continuous setting, one may represent this by a Dirac delta function,

$$p(x,y) = P_0\,\delta(x-x_c)\,\delta(y-y_c). \tag{4}$$

In the discrete formulation (Section 4), this point load is approximated as an equivalent uniform load over the central cell.

2.2 Bending moments and shear forces

In Kirchhoff plate theory the bending and twisting moments per unit length are related to the deflection by

$$M_{xx} = -D\left(\frac{\partial^2 w}{\partial x^2} + \nu\frac{\partial^2 w}{\partial y^2}\right), \tag{5}$$

$$M_{yy} = -D\left(\frac{\partial^2 w}{\partial y^2} + \nu\frac{\partial^2 w}{\partial x^2}\right), \tag{6}$$

$$M_{xy} = -D(1-\nu)\frac{\partial^2 w}{\partial x \partial y}. \tag{7}$$

The Kirchhoff shear resultants are then

$$Q_x = \frac{\partial M_{xx}}{\partial x} + \frac{\partial M_{xy}}{\partial y}, \tag{8}$$

$$Q_y = \frac{\partial M_{yy}}{\partial y} + \frac{\partial M_{xy}}{\partial x}. \tag{9}$$

In the numerical implementation, these derivatives are approximated by central finite differences applied to the discrete field w_{ij}.

3 Boundary Conditions

The plate has three simply supported edges and one clamped edge. Let the coordinate axes be such that $0 \le x \le a$, $0 \le y \le b$, with the bottom edge at $y = 0$ taken as clamped. In the code, the deflection is set to zero on all edges, and the clamped behaviour at $y = 0$ is enforced approximately through a modified finite-difference stencil for the second derivative.

3.1 Simply supported edge

For a simply supported edge with outward normal n, the ideal boundary conditions are

$$w = 0, \tag{10}$$

$$M_n = 0, \tag{11}$$

where

$$M_n = M_{xx} n_x^2 + M_{yy} n_y^2 + 2M_{xy} n_x n_y.$$

3.2 Clamped edge

A clamped edge satisfies

$$w = 0, \tag{12}$$

$$\frac{\partial w}{\partial n} = 0. \tag{13}$$

4 Finite Difference Discretisation

4.1 Grid and unknowns

The plate domain is discretized by a uniform grid with N_x nodes in the x-direction and N_y nodes in the y-direction, including the boundary nodes. For the convergence study, a square grid is used with $N_x = N_y = N$, and

$$h_x = \frac{a}{N_x - 1}, \tag{14}$$

$$h_y = \frac{b}{N_y - 1}. \tag{15}$$

The unknowns of the system are the deflections at interior grid points,

$$w_{ij} \approx w(x_i, y_j), \tag{16}$$

for $i = 1, \dots, N_x - 2$, $j = 1, \dots, N_y - 2$, where $x_i = ih_x$, $y_j = jh_y$. The deflection at the boundary nodes is set to zero.

The interior nodal values are stacked in a column vector $\mathbf{w} \in \mathbb{R}^{n_x n_y}$, where $n_x = N_x - 2$, $n_y = N_y - 2$.

4.2 Discrete Laplacian in 1D

In one spatial dimension, the second derivative is approximated at the interior node i by the central difference

$$\left.\frac{\mathrm{d}^2 w}{\mathrm{d}x^2}\right|_{x_i} \approx \frac{w_{i-1} - 2w_i + w_{i+1}}{h_x^2}. \tag{17}$$

whose first and last rows are modified depending on the boundary type (simply supported or clamped) to approximate the respective edge conditions.

4.3 Biharmonic operator

The biharmonic operator ∇^4 is then approximated by

$$A = LL. \tag{18}$$

Thus the discrete counterpart of the governing equation (2) is

$$DA\mathbf{w} = \mathbf{p}, \tag{19}$$

or, equivalently,

$$A\mathbf{w} = \mathbf{f}, \qquad \mathbf{f} = \frac{\mathbf{p}}{D}. \tag{20}$$

4.4 Approximation of the central point load

The continuous point load P_0 at the plate centre is approximated as a uniform distributed load over the central cell of area $h_x h_y$. The equivalent load per unit area is

$$p_{\text{eff}} = \frac{P_0}{h_x h_y}. \tag{21}$$

This load is applied at the center $(a/2, b/2)$ because N is chosen odd. The corresponding entry in the right-hand side vector is

$$f_c = \frac{p_{\text{eff}}}{D} = \frac{P_0}{D\, h_x h_y}. \tag{22}$$

5 Post-processing: Moments and Shear Forces

Once the deflection field w_{ij} is computed for the full grid (with zero boundary values), second derivatives are approximated using central differences:

$$\left.\frac{\partial^2 w}{\partial x^2}\right|_{i,j} \approx \frac{w_{i-1,j} - 2w_{i,j} + w_{i+1,j}}{h_x^2}, \tag{23}$$

$$\left.\frac{\partial^2 w}{\partial y^2}\right|_{i,j} \approx \frac{w_{i,j-1} - 2w_{i,j} + w_{i,j+1}}{h_y^2}, \tag{24}$$

$$\left.\frac{\partial^2 w}{\partial x \partial y}\right|_{i,j} \approx \frac{w_{i+1,j+1} - w_{i+1,j-1} - w_{i-1,j+1} + w_{i-1,j-1}}{4h_x h_y}. \tag{25}$$

Substituting into (5)–(7) gives discrete fields $M_{xx,ij}$, $M_{yy,ij}$, $M_{xy,ij}$.

The shear forces are then obtained by differentiating the discrete moments using central differences, following (8)–(9), e.g.

$$\left.\frac{\partial M_{xx}}{\partial x}\right|_{i,j} \approx \frac{M_{xx,i+1,j} - M_{xx,i-1,j}}{2h_x}, \tag{26}$$

$$\left.\frac{\partial M_{xy}}{\partial y}\right|_{i,j} \approx \frac{M_{xy,i,j+1} - M_{xy,i,j-1}}{2h_y}, \tag{27}$$

and similarly for the derivatives in y.